

Predicting Customer Churn Using Historical Customer and Service Attributes

Course: MDSDF480 – Advanced Apex Project I
Programme: M.Sc. Data Science and Artificial Intelligence
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1. Executive Summary

Customer churn is a major challenge for telecom companies, as losing existing customers can directly impact business performance and revenue. This project developed a machine learning model to predict customer churn using the **IBM Telco Customer Churn dataset**, which contains customer demographic, service usage, contract, and billing information. The project followed a complete machine learning workflow consisting of data preparation, exploratory data analysis (EDA), feature engineering, model development and performance evaluation.

The dataset was first inspected and cleaned by correcting data types and handling missing values. Exploratory Data Analysis was then performed to understand customer characteristics and identify factors associated with churn. Based on these findings, four engineered features: **TenureCohort**, **MonthlyChargeBand**, **ServiceBundleCount**, and **RevenuePerMonth** were created to better represent customer lifecycle, pricing behaviour and service adoption.

A **Logistic Regression classifier** was selected as the baseline model because it is well suited for binary classification and provides interpretable results. The modelling pipeline included train-test splitting, feature encoding, scaling, cross-validation and hyperparameter tuning using GridSearchCV. During feature engineering, the MonthlyChargeBand feature was reconstructed after the train-test split using quartile boundaries derived only from the training dataset to ensure a leakage-free feature engineering workflow.

The analysis showed that customer churn is associated with customer lifecycle, contract type, service adoption, billing behaviour and pricing characteristics. The final baseline Logistic Regression model achieved a **ROC-AUC of 0.842** and a **recall of approximately 79.4%** on the test dataset. Cross-validation indicated that the model generalised well to unseen data, while hyperparameter tuning did not produce any meaningful improvement. Therefore, the baseline Logistic Regression model was retained as the final model for interpretation and business recommendations.

2. Introduction

2.1 Business Context

Customer churn is an important business challenge for subscription-based industries such as telecommunications. Customers have multiple service providers to choose from, making it relatively easy to switch when they are dissatisfied with pricing, service quality or contract terms. As a result, telecom

companies need effective methods to identify customers who are more likely to discontinue their services so that appropriate retention strategies can be planned.

Customer data generated through routine business operations provides valuable information about customer demographics, subscribed services, billing behaviour and account history. Analysing these attributes using machine learning techniques can help identify patterns associated with customer churn and support data-driven decision-making.

2.2 Problem Statement

The objective of this project was to develop a supervised machine learning model capable of predicting whether a customer is likely to churn based on historical customer information. The analysis used the IBM Telco Customer Churn dataset, which contains demographic details, subscribed services, contract information and billing attributes for 7,043 customers. Customer churn was treated as a binary classification problem, where each customer was classified as either **"Yes" (Churn)** or **"No" (Not Churn)**.

2.3 Project Objectives

The objectives of this project were to:

- prepare and clean the Telco Customer Churn dataset for machine learning
- perform Exploratory Data Analysis (EDA) to identify patterns associated with customer churn
- create meaningful engineered features to better represent customer behaviour
- develop and evaluate a Logistic Regression classification model
- interpret the model results to identify the factors most strongly associated with customer churn
- provide business recommendations based on the findings obtained from the analysis

3. Dataset Overview

The IBM Telco Customer Churn dataset was used throughout this project to develop and evaluate a customer churn prediction model. The dataset contains information for **7043 customers** and includes demographic attributes, subscribed telecom services, account information, billing details and the target variable indicating whether a customer discontinued the service. The dataset represents a typical supervised learning classification problem where historical customer information is used to predict future churn behaviour.

The original dataset consisted of **21 variables**, including the target variable **Churn**. Most features were categorical, describing customer demographics (such as gender, senior citizen status, partner and dependents), subscribed services (such as phone, internet, online security, streaming services and technical support), and account characteristics (including contract type, payment method, tenure, monthly charges and total charges). During preprocessing, the TotalCharges column was converted from text to numeric format before further analysis.

The target variable, **Churn**, is a binary categorical variable with two possible outcomes:

Target Value	Description
Yes	Customer discontinued the telecom service
No	Customer continued using the service

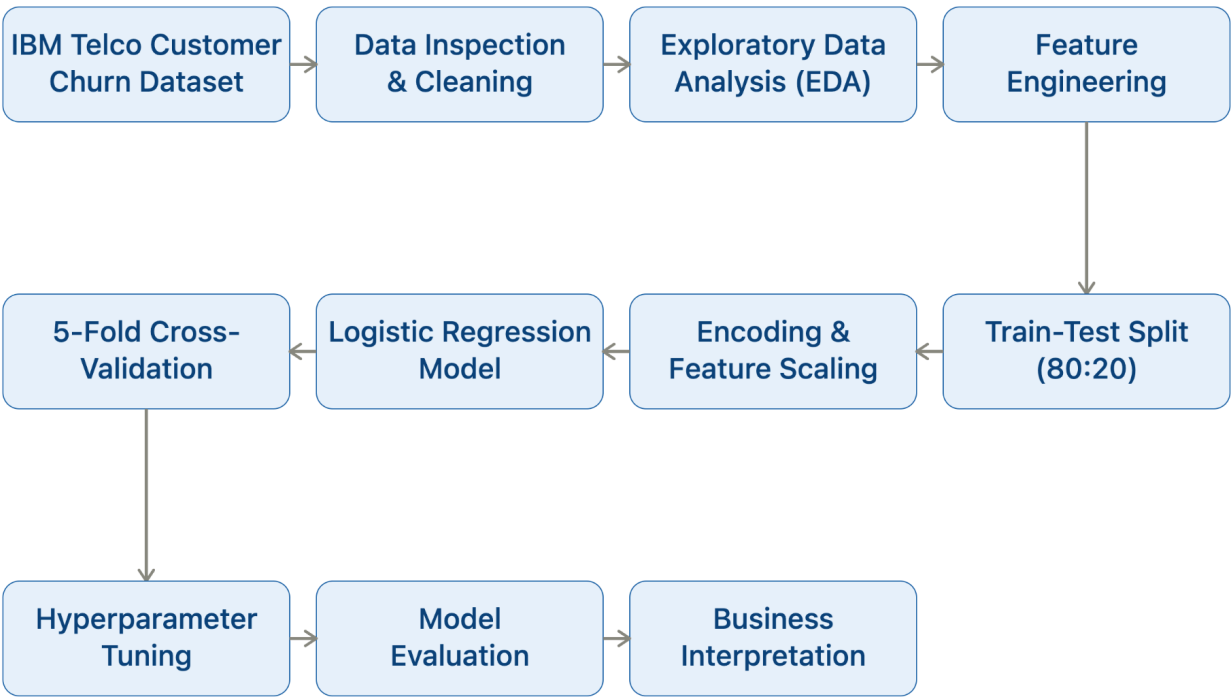
The dataset was used without altering the target labels, allowing the modelling pipeline to directly learn patterns associated with customer churn. Additional features created during the feature engineering stage were incorporated later in the modelling workflow rather than being part of the original dataset.

Attribute	Value
Dataset	IBM Telco Customer Churn
Total Records	7,043
Original Features	21 (including target)
Target Variable	Churn
Problem Type	Binary Classification
Data Types	Numerical and Categorical
Final Engineered Features	TenureCohort, MonthlyChargeBand, ServiceBundleCount, RevenuePerMonth

4. Methodology

This project followed a structured machine learning workflow to develop and evaluate a customer churn prediction model. The workflow began with data inspection and preprocessing, followed by exploratory data analysis (EDA), feature engineering, model development and performance evaluation. Each stage was designed to progressively improve the quality of the dataset and build an interpretable classification model.

The overall methodology adopted in this project is illustrated below:



4.1 Data Preparation

The dataset was initially examined to understand its structure, data types, missing values and overall data quality. During preprocessing, the TotalCharges column was converted from text to a numeric data type, and missing values were handled using domain knowledge because they corresponded to **customers with zero tenure. No duplicate records were identified.** Numerical variables were retained without removing outliers, as these values represented genuine customer behaviour rather than data entry errors.

4.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset and identify patterns associated with customer churn. The analysis examined customer demographics, subscribed services, contract information, billing behaviour, tenure and numerical relationships between variables. Correlation analysis was also carried out to identify potential multicollinearity among numerical features before model development.

4.3 Feature Engineering

Based on the findings from EDA, four additional features were created to capture customer behaviour more effectively:

Engineered Feature	Purpose
TenureCohort	Groups customers according to their relationship duration
MonthlyChargeBand	Categorises customers into monthly billing ranges
ServiceBundleCount	Counts the number of subscribed services
RevenuePerMonth	Represents customer spending behaviour

To maintain a leakage-free modelling workflow, the MonthlyChargeBand feature was reconstructed after the train-test split using quartile boundaries learned only from the training dataset. This ensured that no information from the test data influenced feature creation.

4.4 Model Development

The prepared dataset was divided into training (80%) and testing (20%) subsets using **stratified sampling** to preserve the original class distribution. Categorical variables were encoded using **One-Hot Encoding**, while numerical variables were standardised using **StandardScaler**. These preprocessing steps were integrated into a scikit-learn pipeline to ensure consistent data transformation during training and evaluation. A **Logistic Regression classifier was selected as the baseline model** because of its suitability for binary classification and its interpretability.

4.5 Model Evaluation

Model performance was evaluated using Accuracy, Precision, Recall, F1-score, ROC-AUC, confusion matrix analysis and Receiver Operating Characteristic (ROC) curves. Five-fold cross-validation was used to assess model generalisation, followed by hyperparameter tuning using GridSearchCV. Since **hyperparameter**

optimization did not produce any meaningful improvement over the baseline model, the original Logistic Regression model was retained for interpretation and business recommendations.

5. Exploratory Data Analysis

The purpose of the Exploratory Data Analysis (EDA) was to understand the characteristics of the customer data, identify patterns associated with churn and generate insights that could guide feature engineering and model development. The analysis examined customer demographics, subscribed services, billing information, contract details and numerical relationships between variables. These findings provided the foundation for the engineered features and the subsequent predictive modelling process.

5.1 Customer Churn Distribution

The target variable was first analysed to understand the distribution of customers who churned and those who remained with the telecom service. The dataset exhibited class imbalance, with a substantially larger proportion of customers belonging to the "No Churn" category. Although the imbalance was moderate, **stratified train-test splitting was adopted during model development to preserve the original class distribution** in both datasets. This ensured that the evaluation results remained representative of the original data.

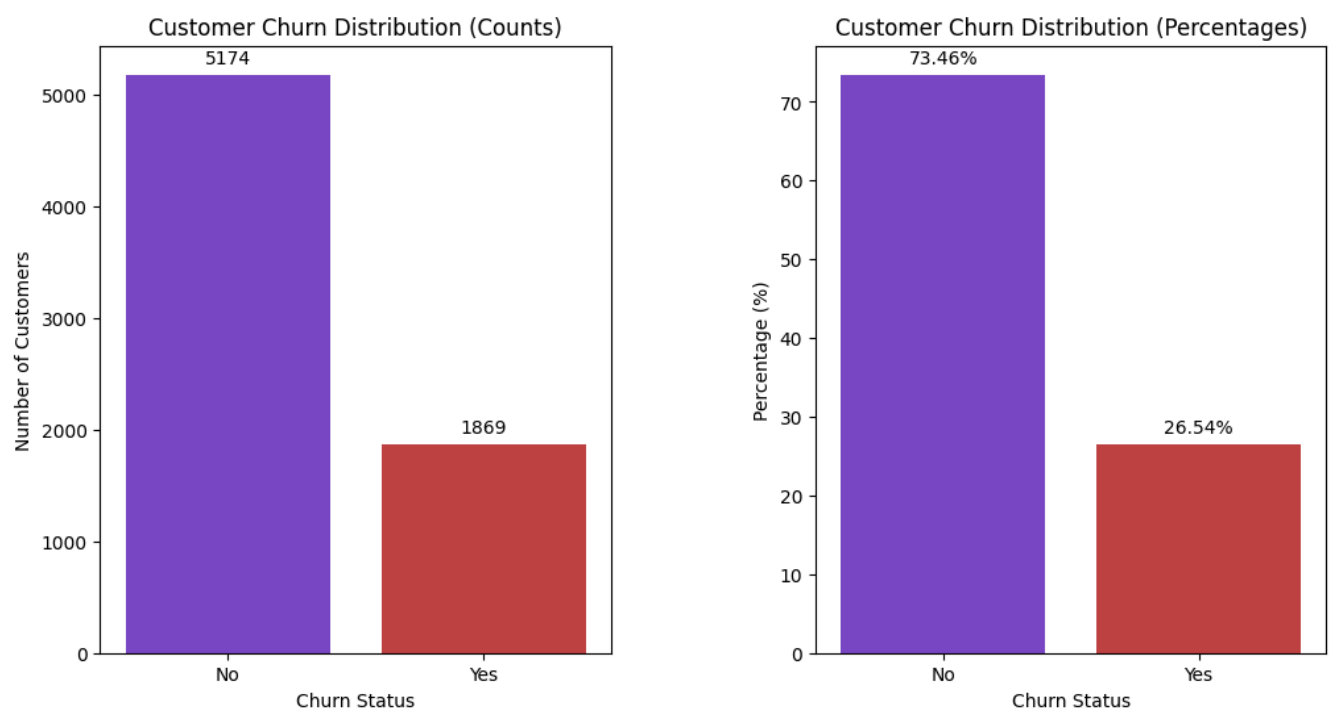


Figure 5.1: Distribution of Customer Churn

5.2 Customer Demographics

Customer demographic variables such as senior citizen status, partner status and dependents were analysed to understand their relationship with churn. The analysis indicated that churn patterns varied across demographic groups, suggesting that customer characteristics influence retention behaviour. However, demographic attributes alone were insufficient to explain churn, highlighting the importance of considering service usage and account-related variables together.

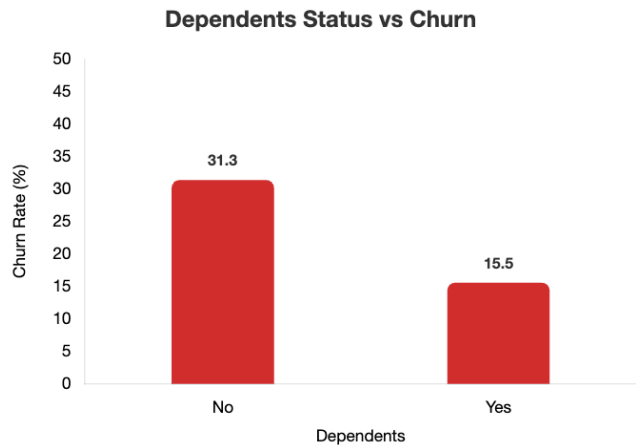
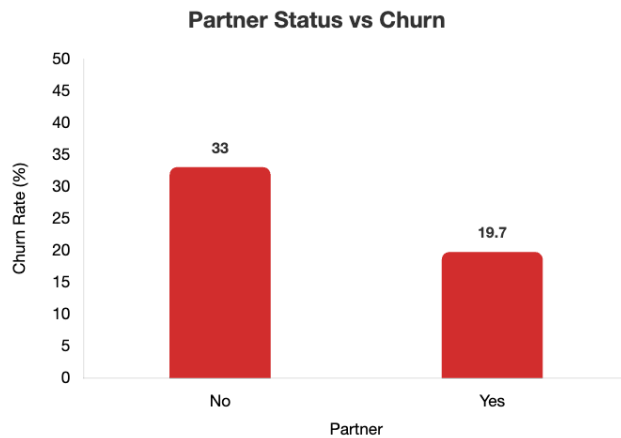
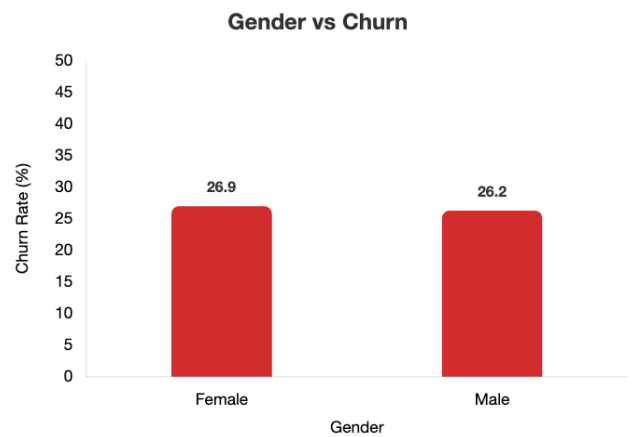
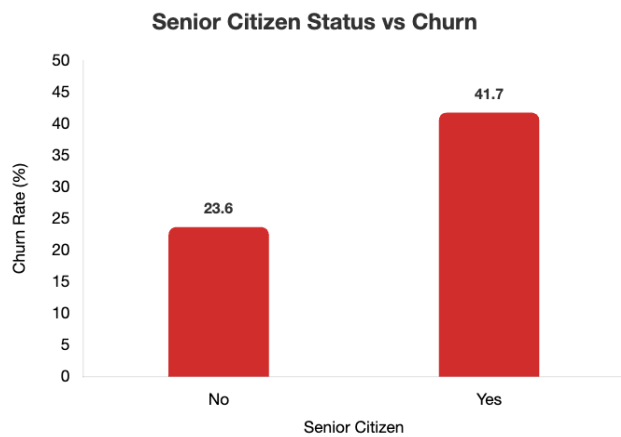


Figure 5.2: Churn by Customer Demographics

5.3 Service Adoption and Contract Characteristics

The analysis of subscribed services and account information revealed stronger associations with churn than demographic variables. Customers with month-to-month contracts, limited service adoption, and fewer value-added services exhibited higher churn rates compared with customers using long-term contracts and multiple bundled services. These observations motivated the creation of engineered features representing customer tenure and service adoption.

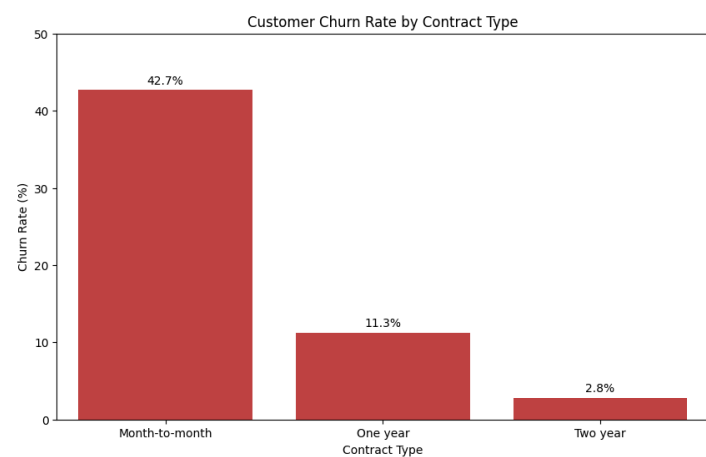
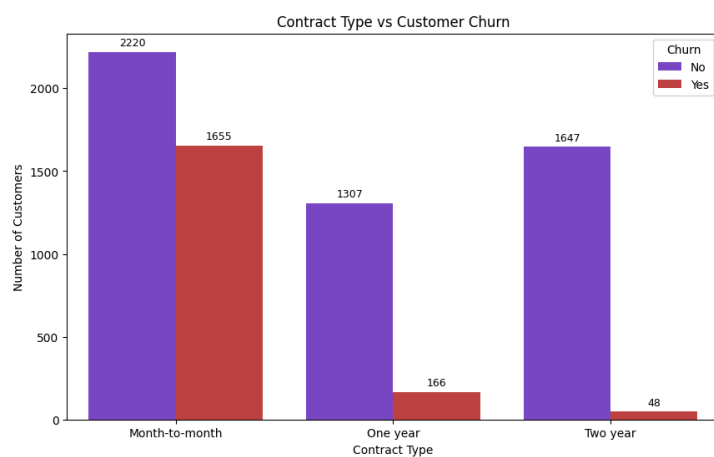


Figure 5.3: Churn by Contract Type

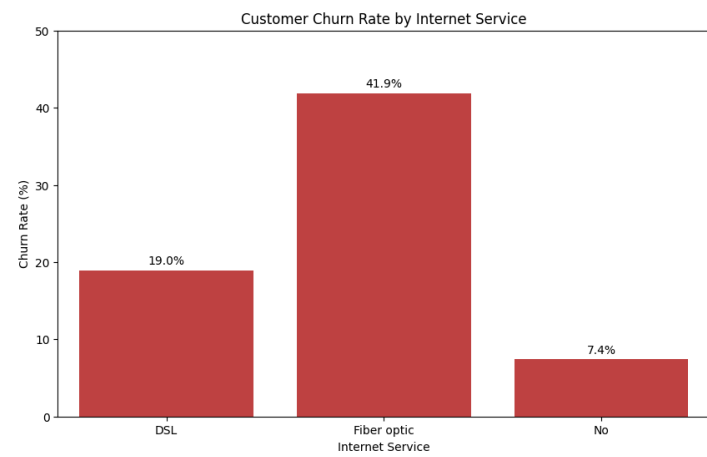
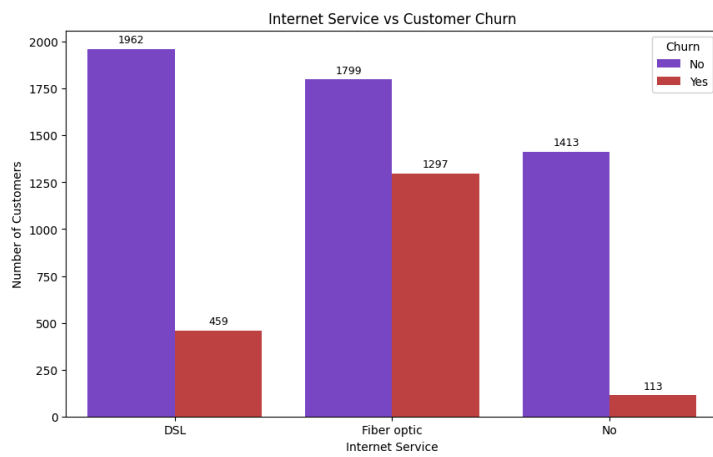


Figure 5.4: Churn by Internet and Service Adoption

5.4 Billing Behaviour

Billing-related variables were examined using numerical distributions and comparative visualisations. **Customers** with shorter tenure generally showed lower cumulative charges, while monthly charges varied across customer segments. These observations suggested that billing characteristics and customer lifecycle could provide useful predictive information, leading to the creation of features such as **MonthlyChargeBand** and **RevenuePerMonth**.

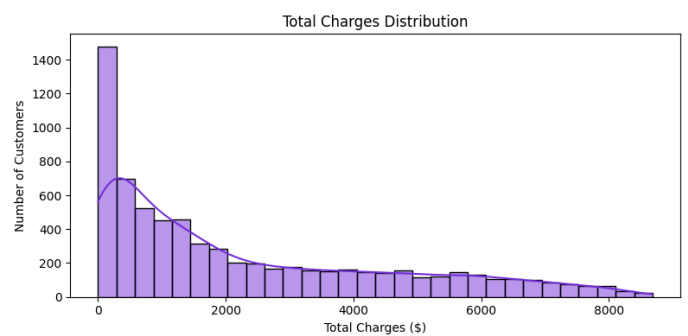
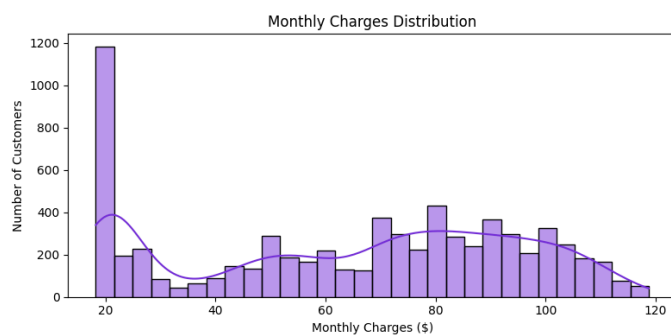


Figure 5.5: Distribution of Monthly Charges

Figure 5.6: Distribution of Total Charges

5.5 Correlation Analysis

Correlation analysis was performed for the numerical variables to understand their relationships and identify potential multicollinearity. A strong positive relationship was observed between tenure and TotalCharges, while MonthlyCharges also showed a moderate positive relationship with TotalCharges. Since the objective of this project was predictive modelling rather than causal inference, these variables were retained because they contributed complementary information to the classification model.

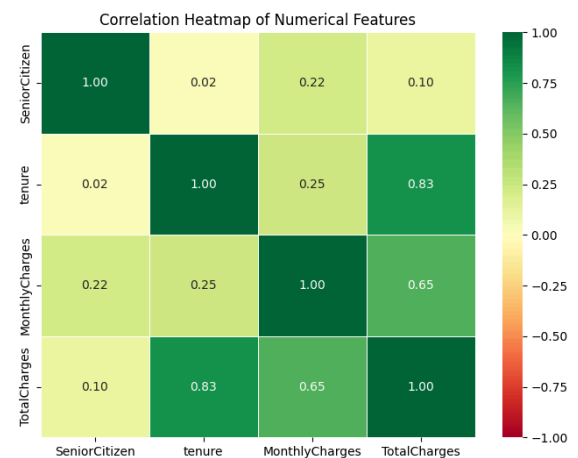


Figure 5.7: Correlation Heatmap

5.6 Key EDA Findings

The exploratory analysis identified several important patterns that informed the feature engineering and modelling stages:

- Customer churn was associated with customer lifecycle, contract type, service adoption, billing behaviour, and pricing characteristics.
- Contract-related variables demonstrated stronger relationships with churn than demographic attributes.
- Numerical correlation analysis identified expected relationships between tenure and cumulative billing variables.
- The insights obtained from EDA directly guided the creation of four engineered features used in the predictive model.

6. Feature Engineering

Feature engineering was performed to create additional variables that could better represent customer behaviour than the original attributes alone. The engineered features were developed based on insights obtained during Exploratory Data Analysis (EDA) and domain understanding of customer lifecycle, service adoption, and billing behaviour. These new features were incorporated into the modelling dataset after validating that they were meaningful and free from data leakage.

6.1 Engineered Features

Four new features were created during this project to enhance the predictive capability of the baseline dataset.

Feature	Description	Motivation
TenureCohort	Groups customers according to their tenure.	Represents different stages of the customer lifecycle.
MonthlyChargeBand	Categorises customers into billing ranges using quartiles.	Captures pricing behaviour while reducing sensitivity to individual values.
ServiceBundleCount	Counts the number of subscribed telecom services.	Measures the overall level of service adoption.
RevenuePerMonth	Represents customer spending behaviour based on billing information.	Provides an additional view of customer value.

These engineered variables complement the original dataset by capturing broader behavioural patterns rather than relying only on the raw input variables.

6.2 Leakage-Free Feature Engineering

An important improvement made during the project involved the creation of the **MonthlyChargeBand** feature.

The feature was initially generated using the complete dataset. However, because the quartile boundaries were calculated from all observations, the feature engineering process was revised before model training. After performing the train-test split, quartile boundaries were learned only from the training dataset and then applied to both the training and testing datasets. This ensured that no information from the test data influenced feature creation and maintained a leakage-free modelling workflow.

This refinement strengthens the validity of the evaluation results by ensuring that the test dataset remained completely unseen during feature engineering.

6.3 Feature Contribution

The engineered features were designed to represent customer behaviour from different perspectives.

- **TenureCohort** captures customer lifecycle stages.
- **MonthlyChargeBand** summarises pricing behaviour.
- **ServiceBundleCount** represents the breadth of subscribed services.
- **RevenuePerMonth** provides an additional billing-related indicator.

Together with the original customer attributes, these engineered variables formed the final modelling dataset used for training and evaluation.

6.4 Summary

Feature engineering extended the original IBM Telco Customer Churn dataset by introducing variables that better represented customer lifecycle, pricing behaviour, and service adoption. The engineered features were created using insights obtained from EDA and incorporated into a leakage-free modelling workflow. These variables formed the final feature set used by the Logistic Regression model for customer churn prediction.

7. Model Development and Evaluation

The prepared dataset was used to develop a supervised machine learning model for customer churn prediction. A Logistic Regression classifier was selected as the baseline model because it is well suited for binary classification problems and produces interpretable coefficients. The modelling pipeline incorporated data preprocessing, cross-validation, hyperparameter tuning, and performance evaluation to ensure a systematic assessment of model performance.

7.1 Model Development

The final modelling dataset included both the original customer attributes and the four engineered features developed during feature engineering. The data was divided into training and testing subsets using an 80:20 stratified split to preserve the original class distribution.

Categorical variables were transformed using One-Hot Encoding, while numerical variables were standardised using **StandardScaler**. These preprocessing steps were combined within a scikit-learn pipeline to ensure that identical transformations were applied during both training and testing.

7.2 Cross-Validation and Hyperparameter Tuning

To evaluate the robustness of the baseline model, five-fold cross-validation was performed on the training dataset. The model was subsequently optimised using GridSearchCV to identify suitable hyperparameter values.

The cross-validation results demonstrated consistent model performance across different folds, indicating good generalisation capability. Hyperparameter tuning produced only marginal changes in the evaluation metrics and did not provide a meaningful improvement over the baseline model. Consequently, the original Logistic Regression model was retained for the final analysis and interpretation.

7.3 Model Performance

The final Logistic Regression model was evaluated using multiple classification metrics to obtain a balanced assessment of predictive performance.

Metric	Accuracy	Precision	Recall	F1-score	ROC-AUC
Result	0.734	0.499	0.794	0.613	0.842

Table 7.1: Performance Metrics of the Final Logistic Regression Model.

Rather than relying on a single metric, the evaluation considered multiple complementary measures. Recall was particularly relevant because correctly identifying customers who are likely to churn is an important objective of churn prediction. The ROC-AUC score of **0.842** indicates that the model achieved good discrimination between customers who churned and those who remained.

7.4 Confusion Matrix Analysis

The confusion matrix was used to examine the model's classification performance in greater detail by comparing predicted and actual customer outcomes.

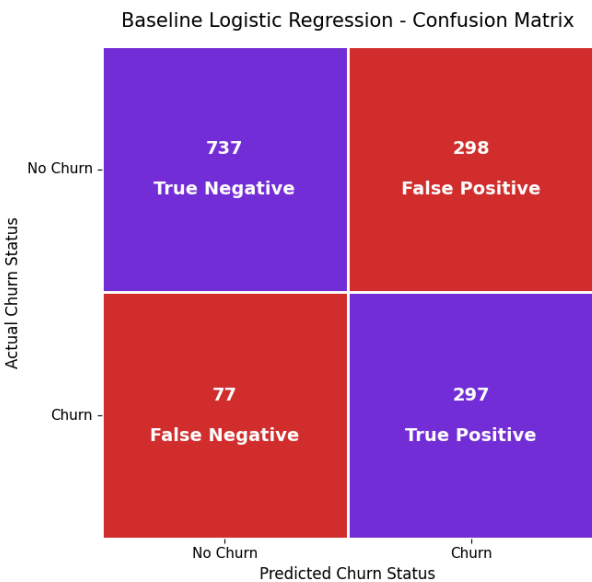


Figure 7.1: Confusion Matrix

The confusion matrix showed that the model correctly identified the majority of customers in both classes while maintaining a relatively high recall for churn prediction. As with most binary classification models, some false positives and false negatives remained, illustrating the trade-off between correctly identifying churning customers and avoiding incorrect churn predictions.

7.5 Model Interpretation

One advantage of Logistic Regression is that it provides interpretable coefficients that help explain how different variables influence the model's predictions.

The coefficient analysis indicated that both the original customer attributes and the engineered features contributed to churn prediction. Variables related to customer tenure, contract characteristics, service adoption, and billing behaviour exhibited stronger relationships with the predicted probability of churn, which was consistent with the patterns observed during EDA.

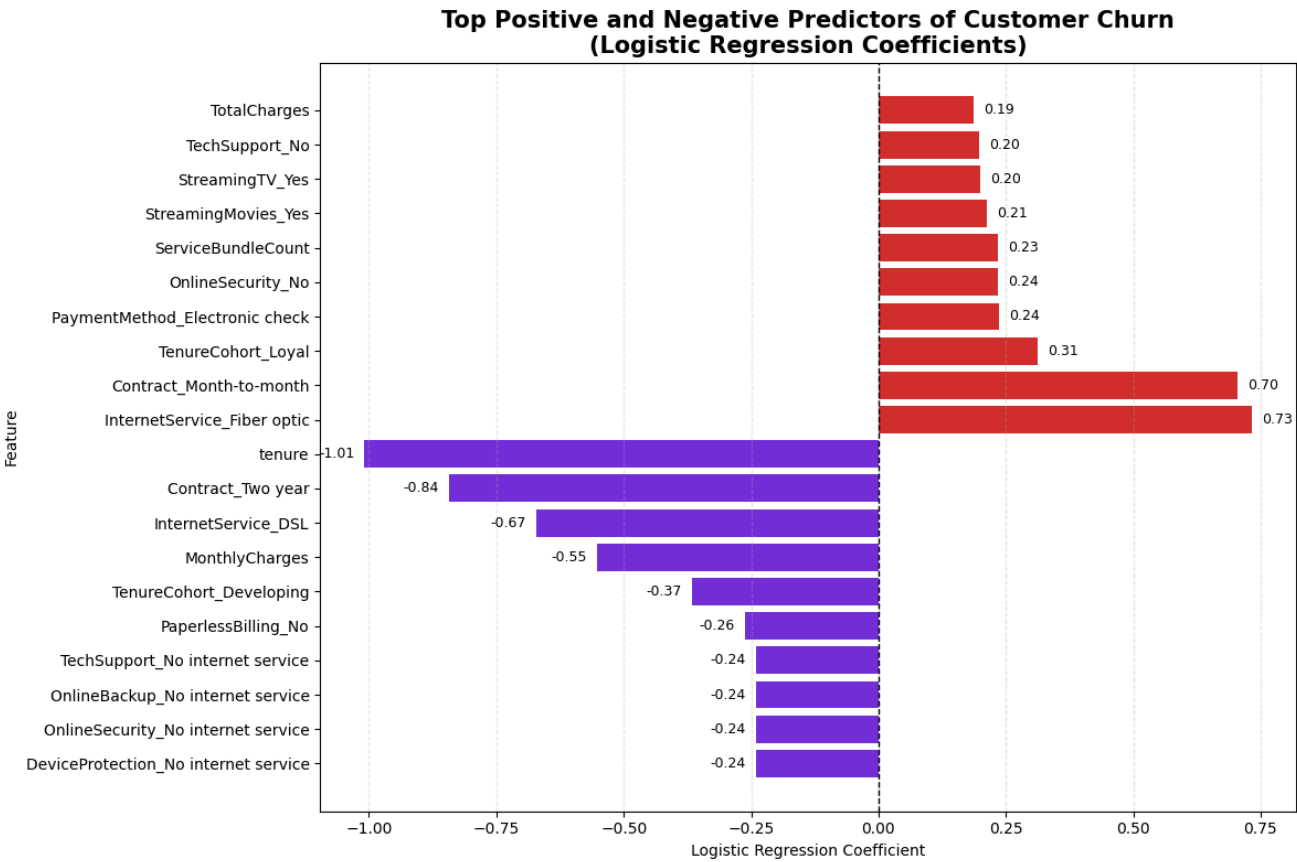


Figure 7.2: Logistic Regression Coefficient Plot

Multicollinearity Consideration

Correlation analysis performed during EDA identified a strong positive relationship between tenure and TotalCharges, with MonthlyCharges also showing a moderate positive relationship with TotalCharges. Since the primary objective of this project was prediction rather than causal inference, these variables were retained because they provided complementary predictive information. Although multicollinearity can influence the magnitude of individual coefficients, it does not necessarily reduce the predictive performance of a regularised Logistic Regression model.

7.6 Key Findings

The modelling stage produced the following observations:

- The Logistic Regression baseline model demonstrated good predictive performance on the Telco Customer Churn dataset.
- Cross-validation indicated that the model generalised consistently across different training folds.
- Hyperparameter tuning did not produce a meaningful improvement over the baseline model.
- The engineered features complemented the original dataset and formed part of the final modelling pipeline.

The model identified customer lifecycle, contract characteristics, service adoption, and billing behaviour as important contributors to churn prediction.

8. Business Recommendations

The exploratory analysis and the final Logistic Regression model identified customer lifecycle, contract type, service adoption, billing behaviour, and pricing characteristics as the primary factors associated with customer churn. Based on these findings, the following recommendations can help support customer retention strategies.

8.1 Strengthen Customer Retention During Early Tenure

Customers with shorter tenure exhibited a greater tendency to churn than long-term customers. This suggests that the initial stages of the customer relationship are particularly important for customer retention.

Telecom providers should focus on improving customer onboarding, providing timely support, and maintaining regular engagement during the early months of the customer lifecycle. These measures may improve customer satisfaction and encourage longer-term retention.

8.2 Encourage Long-Term Contract Adoption

The analysis indicated that contract type was strongly associated with customer churn, with month-to-month customers showing higher churn than customers with longer-term contracts.

Offering attractive incentives for annual or multi-year contracts may encourage customers to remain with the service provider for longer periods while reducing churn risk.

8.3 Promote Value-Added Services

Service adoption was another important factor identified during the analysis. Customers using a broader range of services generally demonstrated different churn behaviour compared with customers subscribing to fewer services.

Encouraging customers to adopt relevant value-added services may increase customer engagement and strengthen long-term relationships with the service provider.

8.4 Monitor Billing and Pricing Behaviour

Billing-related variables, including monthly charges and engineered pricing features, contributed to churn prediction. Monitoring customers experiencing pricing changes or unusual billing patterns may help identify customers requiring additional engagement before they decide to discontinue the service.

8.5 Integrate Predictive Analytics into Decision-Making

The developed Logistic Regression model demonstrated good predictive performance and provides interpretable results that can support customer retention initiatives.

Rather than replacing existing business processes, the model can serve as a decision-support tool by helping prioritise customers who may benefit from proactive retention efforts.

9. Limitations

- The analysis was performed using a single publicly available dataset and therefore reflects only the characteristics captured within that dataset.
- Logistic Regression was intentionally selected as the baseline classification model. More complex machine learning algorithms were not explored within the scope of this project.
- The engineered features were developed using the available customer attributes. Additional behavioural or temporal data may further improve predictive performance.
- The model was evaluated using historical data and was not validated in a real-time production environment.

10. Conclusion

This project developed an end-to-end machine learning pipeline for predicting customer churn using the IBM Telco Customer Churn dataset. The workflow included data preparation, exploratory data analysis, feature engineering, model development, and performance evaluation using a Logistic Regression classifier. Four engineered features were created to better represent customer lifecycle, pricing behaviour, and service adoption, while a leakage-free feature engineering workflow was maintained throughout the modelling process.

The final model achieved a ROC-AUC of 0.842 and a recall of approximately 79.4%, demonstrating good predictive performance on the test dataset. Cross-validation indicated consistent generalisation across training folds, while hyperparameter tuning did not produce a meaningful improvement over the baseline model. These findings supported retaining the original Logistic Regression model for interpretation and business recommendations.

Overall, the project demonstrates how data preprocessing, exploratory analysis, and carefully designed feature engineering can improve the development of an interpretable customer churn prediction model. The resulting workflow provides a structured approach that can support future customer retention analyses and serves as a strong foundation for extending the model using more advanced machine learning techniques.
